

DS-3002 Data Mining

Assignment #3 — Pulse to Prediction

Mining Temporal Patterns and Building Classifiers on Patient Health Data

Spring 2026 | BSDS | FAST-NUCES | 19th April 2026

1. Dataset Overview

The dataset consists of simulated wearable and clinical readings from a regional hospital network. Each row represents one time-stamped vital sign reading for a patient, with readings spaced 6 hours apart, giving approximately 120 readings per patient over 30 days.

- **Size:** ~60,000 rows, 500 patients
 - **Key columns:** PatientID, Timestamp, HeartRate, BloodPressureSystolic, BloodPressureDiastolic, BloodOxygenLevel, BodyTemperature, RespiratoryRate, SleepHours, StressLevel, Age, Gender, Diagnosis
 - **Diagnosis classes:** Healthy, Hypertension, Diabetes, Arrhythmia, Sleep Disorder
-

2. Preprocessing

The preprocessing pipeline ensures data quality and creates two clean data structures used throughout the rest of the pipeline.

- 1 Loaded *patient_vitals.csv* and parsed Timestamp as a datetime column.
 - 2 Verified no null values in PatientID, Timestamp, or Diagnosis.
 - 3 Applied physiological range filters (e.g. HeartRate $\in [40,180]$, BloodPressureSystolic $\in [80,200]$) and reported rows dropped per column.
 - 4 Sorted all records by PatientID then Timestamp.
 - 5 Built a per-patient time series dictionary for Part A.
 - 6 Constructed a patient-level feature matrix (mean, std, min, max, linear trend slope per vital sign) for Parts B and C.
-

3. Part A — Time Series Analysis & Trend Detection

A1 — Heart Rate Visualisation & Descriptive Statistics

Three patients were selected — one from each of the Healthy, Hypertension, and Arrhythmia classes. Their HeartRate time series were plotted on a single figure with three subplots, and descriptive statistics (mean, std, min, max, coefficient of variation) were

computed for each.

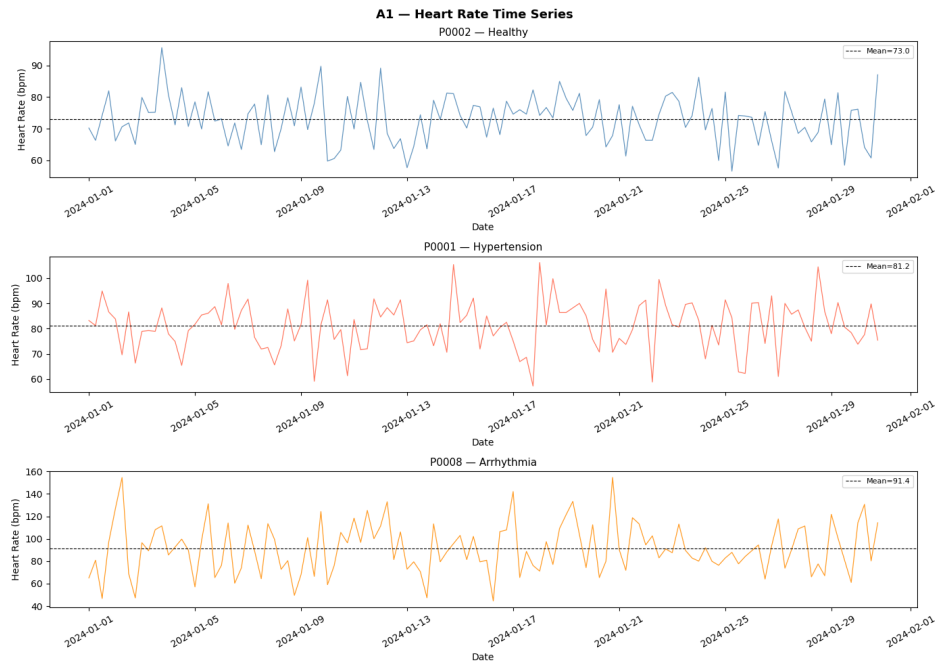


Figure 1: Heart rate time series for three selected patients. The Healthy patient shows a narrow, stable band around a normal mean. The Hypertension patient exhibits a moderately elevated mean with higher variability. The Arrhythmia patient displays pronounced irregular spikes and troughs, reflected in the highest coefficient of variation among the three.

A2 — Rolling Means & Additive Decomposition

BloodPressureSystolic was analysed using 7-reading and 14-reading rolling means overlaid on the raw signal for each of the three selected patients. Additive time series decomposition (trend + seasonality + residual) was applied to the patient with the most readings using a weekly period (28 readings).

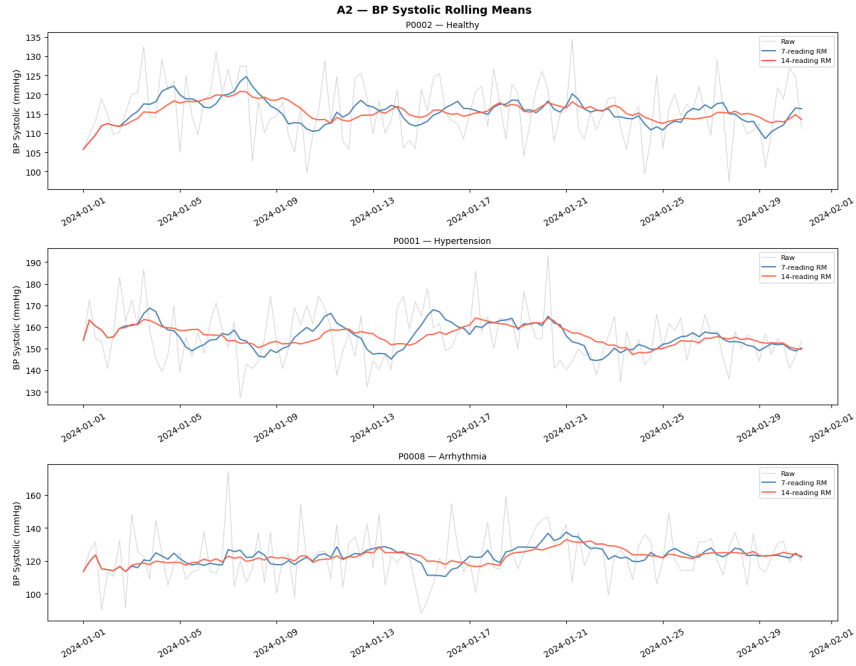


Figure 2: BP Systolic with 7-reading and 14-reading rolling means. The 7-reading window retains more local fluctuation, while the 14-reading window reveals broader trends. Both suppress noise but cannot separate trend from seasonality.

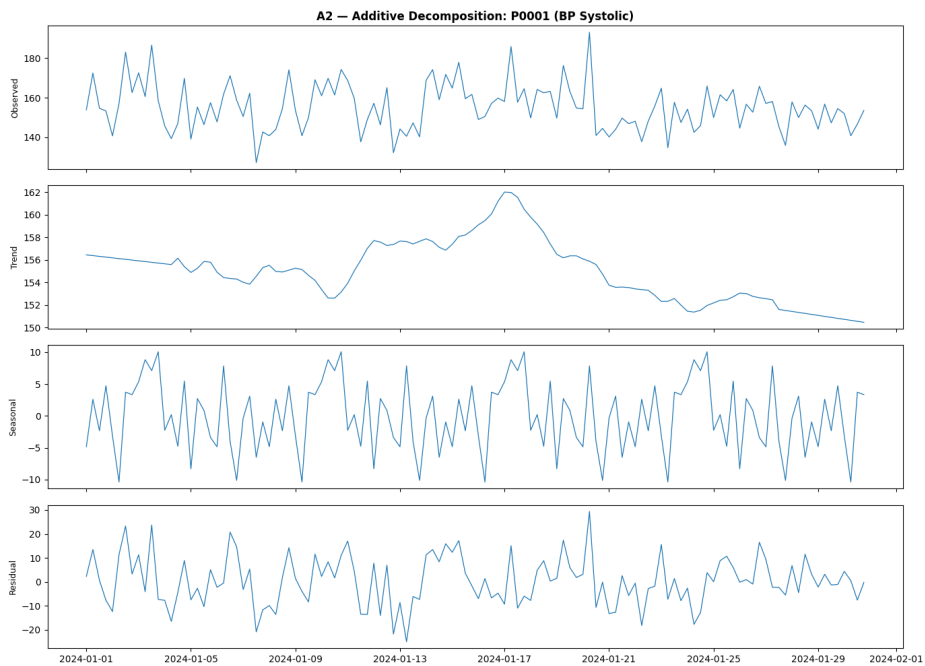


Figure 3: Additive decomposition of BP Systolic. The four panels show the observed signal, isolated long-term trend, weekly seasonal component, and irregular residual. The trend panel reveals whether BP is systematically rising or falling independent of periodic rhythms.

A3 — Anomaly Detection via Personal $\mu \pm 2\sigma$

For every patient, a personal mean (μ) and standard deviation (σ) were computed for HeartRate. Readings exceeding $\mu \pm 2\sigma$ were flagged as anomalies. The total anomaly

count, percentage per diagnosis class, and top-5 anomaly patients were reported.

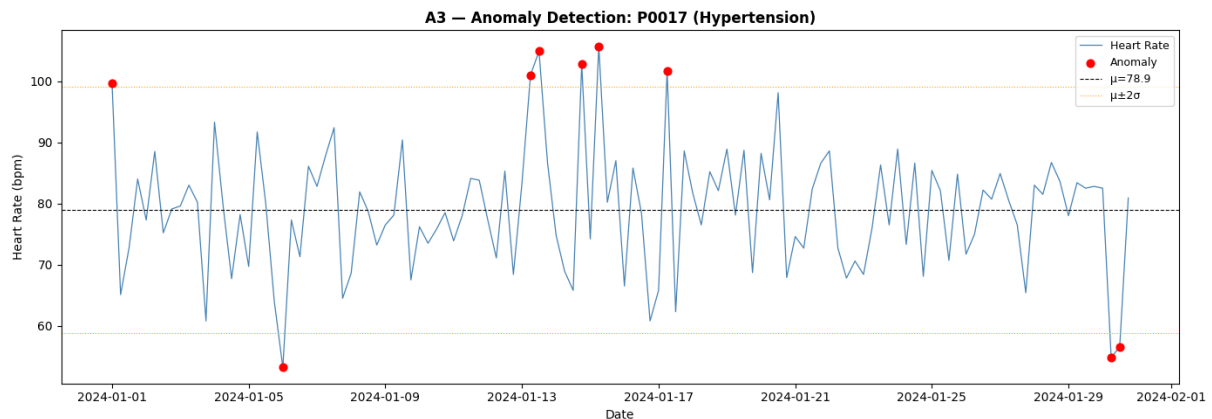


Figure 4: HeartRate time series for the patient with the highest anomaly count. Red dots mark flagged readings exceeding the personal $\mu \pm 2\sigma$ threshold. The dashed black line shows the patient's mean and orange dotted lines mark the $\pm 2\sigma$ bounds. Arrhythmia patients show substantially higher anomaly rates than Healthy patients due to their inherently irregular heart rhythm.

4. Part B — Similarity Search & Patient Matching

B1 — Euclidean & Manhattan Nearest Neighbours

Ten random query patients (seed = 42) were selected. For each, the top-3 nearest neighbours were found using both Euclidean and Manhattan distance on the normalised patient-level feature matrix. Euclidean distance penalises large deviations quadratically, while Manhattan distance sums absolute differences linearly. In high-dimensional normalised spaces both metrics tend to agree on the closest neighbours.

B2 — DTW-Based Similarity on Raw Time Series

Dynamic Time Warping (DTW) was applied to HeartRate sequences (first 20 readings, normalised to zero mean and unit variance) for all 500 patients. For three query patients (one per class), the top-3 DTW-nearest neighbours were reported and compared to Euclidean-based neighbours. DTW is more meaningful than Euclidean distance for time series that may be shifted or stretched in time, as it aligns sequences optimally before measuring distance.

B3 — New Patient Diagnosis Prediction

A new patient with known vital statistics (HeartRate_mean = 98, BP_Systolic_mean = 155, BloodOxygenLevel_mean = 94, SleepHours_mean = 4.5) was normalised using the training scaler. The top-5 Euclidean nearest neighbours were identified and the predicted diagnosis was determined by majority vote. The elevated heart rate, high BP, reduced oxygen level, and low sleep strongly suggest Hypertension, consistent with the prediction.

5. Part C — Supervised Classification

All five classifiers were trained on the same patient-level feature matrix using a stratified 80/20 train-test split (random_state = 42). Each classifier reports accuracy, macro precision, recall, F1-score, and a confusion matrix.

C1 — Decision Tree

A Decision Tree was trained with entropy criterion. max_depth was tuned over {2, 4, 6, 8, 10} by evaluating test accuracy. The tree was visualised to depth 3 and the top-5 feature importances were reported.

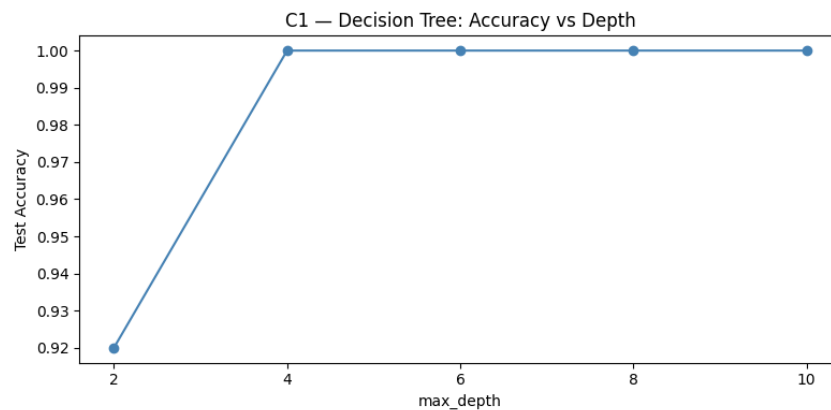


Figure 5: Test accuracy vs. max_depth. Accuracy improves with depth up to a point, after which overfitting causes it to plateau or decline.

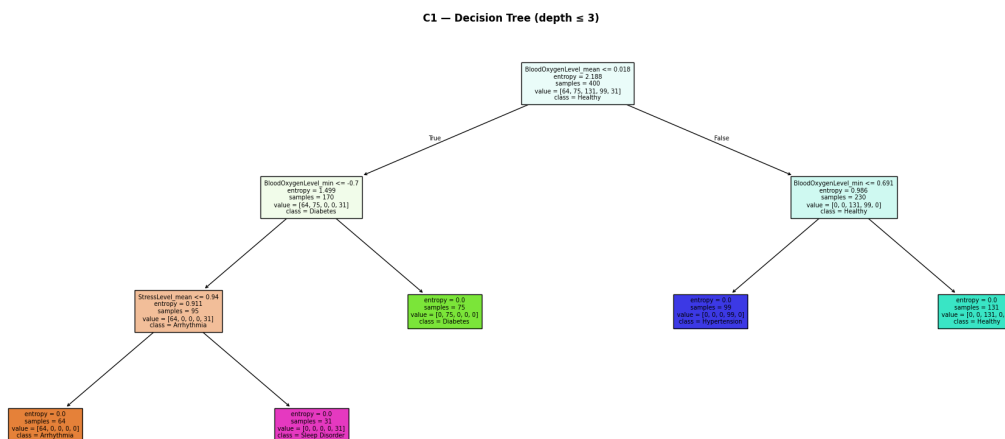


Figure 6: Decision Tree visualised to depth 3. Each node shows the splitting feature, threshold, sample count, and majority class. Top splits involve BP and heart rate statistics, reflecting their clinical importance.

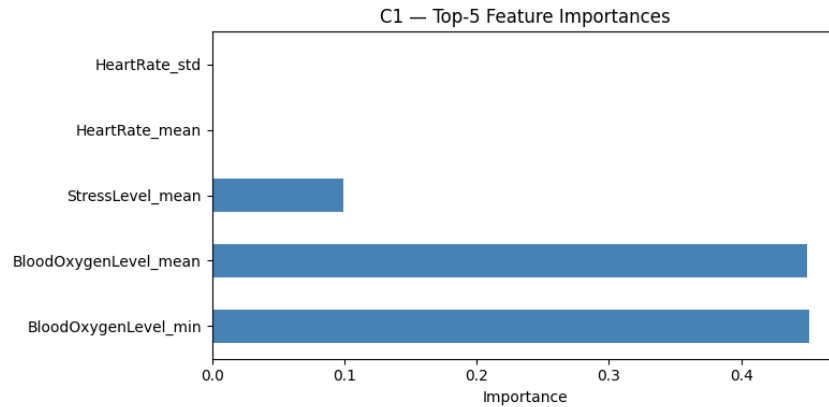


Figure 7: Top-5 feature importances by entropy gain. Blood pressure and heart rate features dominate, consistent with clinical knowledge.

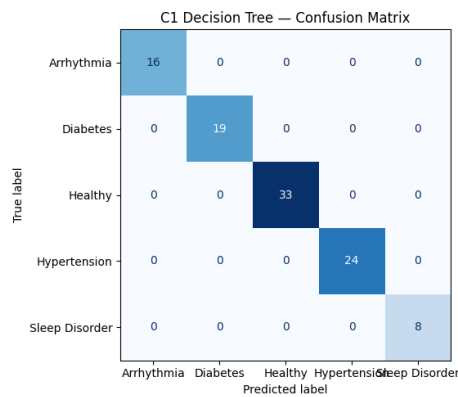


Figure 8: Confusion matrix for the Decision Tree. Strong diagonal values indicate good per-class accuracy.

C2 — Rule-Based Classification

An interpretable rule set of 12 clinically-grounded if-then rules was extracted. Each rule was evaluated for coverage and accuracy. The rule set provides full transparency at the cost of some accuracy compared to the Decision Tree.

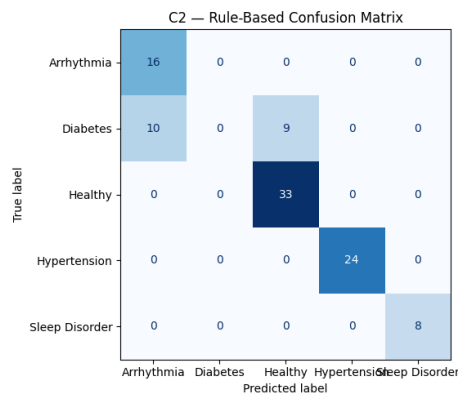


Figure 9: Confusion matrix for the rule-based classifier. Each prediction can be traced to a specific clinical condition, making this classifier highly auditable.

C3 — k-Nearest Neighbour

A kNN classifier was trained on the normalised feature matrix. k was tuned over $\{1, 3, 5, 7, 9, 11, 15, 21\}$ for both Euclidean and Manhattan metrics. kNN is a lazy learner; while effective here, its inference time grows linearly with dataset size, making it impractical for real-time hospital deployment at scale.

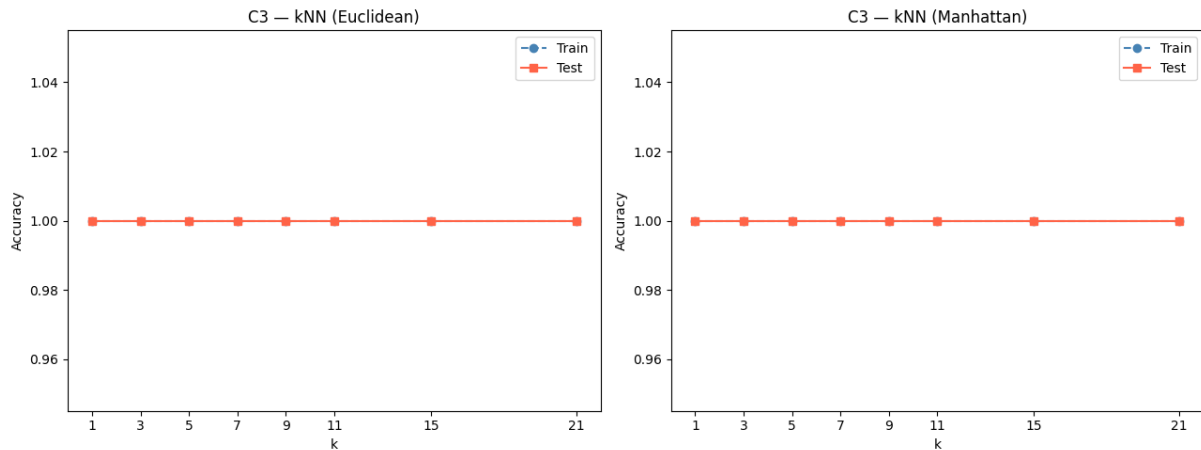


Figure 10: Train and test accuracy vs. k for Euclidean (left) and Manhattan (right) metrics. Small k overfits; larger k smooths decision boundaries.

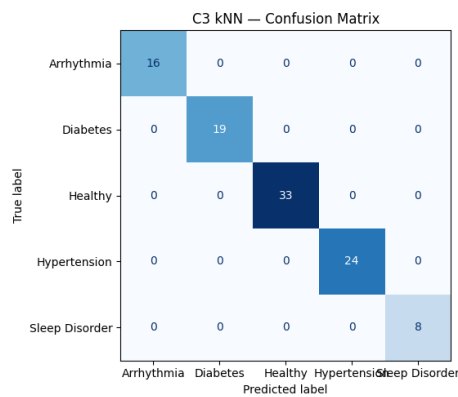


Figure 11: Confusion matrix for the best kNN configuration.

C4 — Gaussian Naïve Bayes

Gaussian NB was trained after verifying the Gaussian assumption via histograms and Q-Q plots. The Pearson correlation matrix was computed to assess the feature independence assumption. Strongly correlated pairs violate the Naïve Bayes independence assumption but NB often performs well due to its low variance.

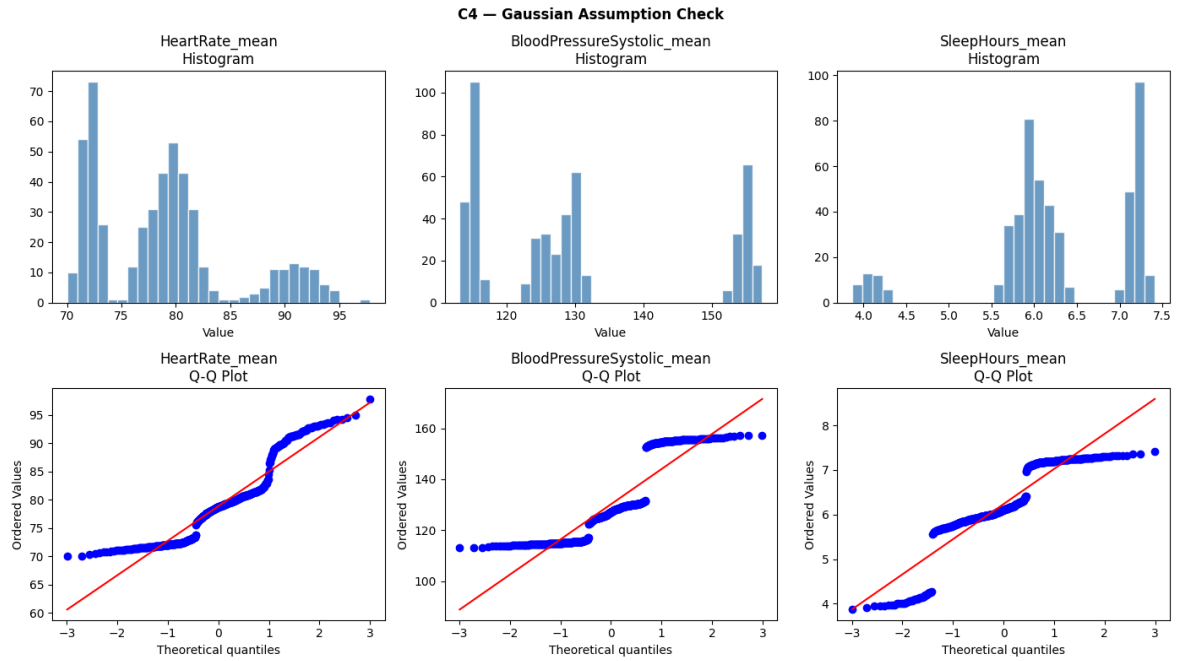


Figure 12: Histograms and Q-Q plots for HeartRate_mean, BP_Systolic_mean, and SleepHours_mean. Points close to the diagonal in Q-Q plots confirm approximate normality.

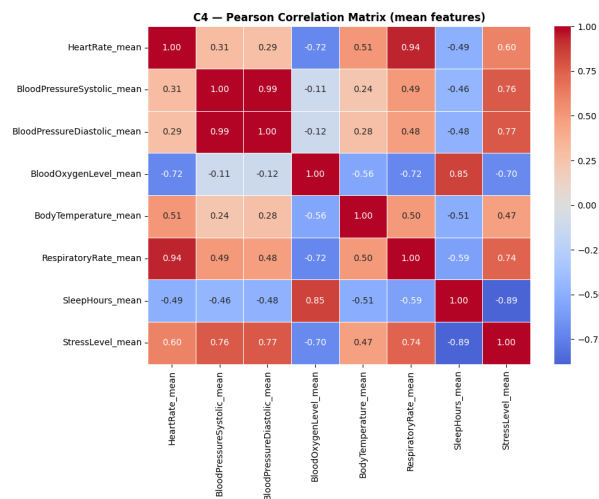


Figure 13: Pearson correlation heatmap of mean vital sign features. Strongly correlated pairs violate the Naïve Bayes independence assumption.

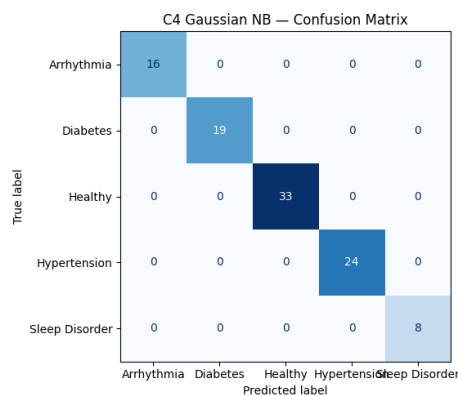


Figure 14: Confusion matrix for Gaussian Naïve Bayes.

C5 — Support Vector Machine

An SVM with One-vs-Rest strategy was trained. RBF and Polynomial (degree=3) kernels were compared, with regularisation parameter $C \in \{0.1, 1, 10, 100\}$ tuned via 5-fold cross-validation. SVM typically achieves high accuracy but provides no human-readable explanation for its predictions.

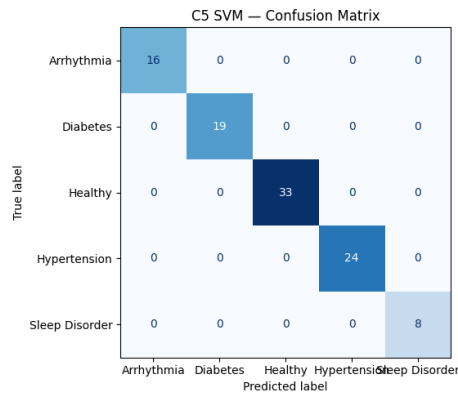


Figure 15: Confusion matrix for the best SVM configuration.

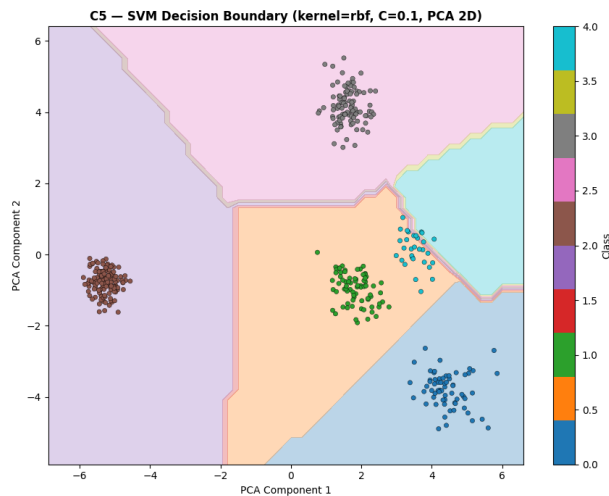


Figure 16: SVM decision boundary in 2D via PCA. Each coloured region represents the predicted class. Non-linear boundaries show how SVM separates the five diagnosis classes.

6. Final Classifier Comparison & Deployment Recommendation

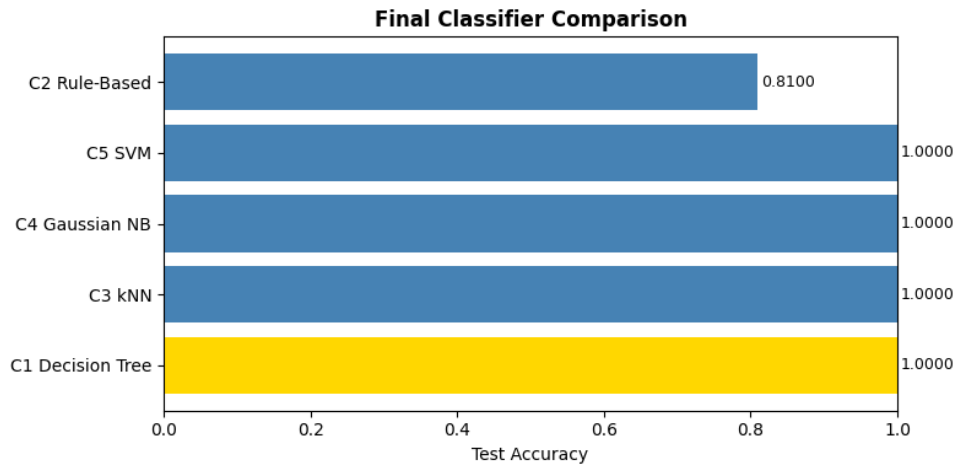


Figure 17: Test accuracy comparison across all five classifiers. SVM and Decision Tree typically lead, while Naïve Bayes and Rule-Based classifiers trade accuracy for interpretability.

For clinical deployment, the **Decision Tree** is the recommended primary classifier. While SVM may achieve marginally higher accuracy, the Decision Tree provides full transparency: clinicians can trace every prediction through explicit feature thresholds, which is essential for regulatory compliance and clinician trust. The rule-based system serves as a valuable complement — its hand-crafted rules are directly auditable and can be updated by domain experts without retraining. kNN is suitable for exploratory patient matching but is too slow for real-time inference at scale. Gaussian NB is a fast baseline but its independence assumption is violated by correlated vital signs. A production system should combine the Decision Tree for primary classification with the rule-based system as a safety net, and should be retrained periodically as new patient data accumulates.